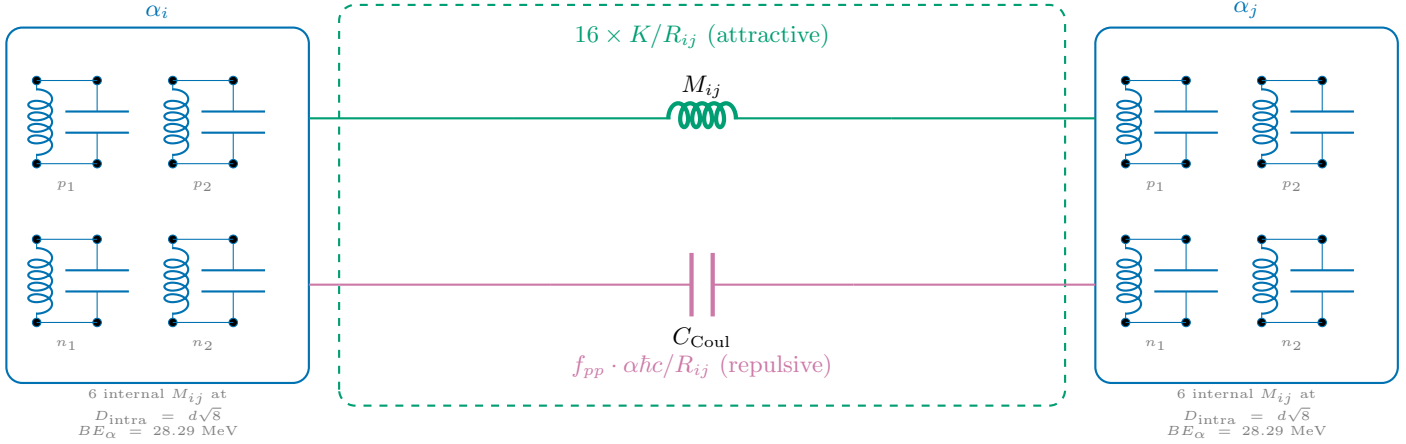


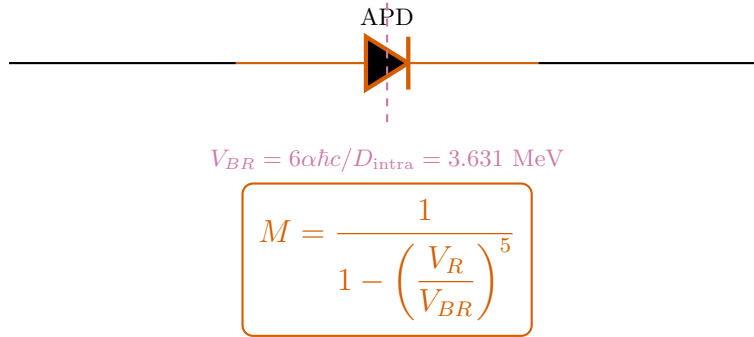
F-19 Semiconductor Equivalent Circuit

$$4\alpha + {}^3\text{H Core+Halo} \quad \text{---} \quad V_R/V_{BR} = 0.030 \quad \text{---} \quad M = 1.000 \text{ (Core+Halo)}$$

A: Inter-Alpha Junction (Unit Cell)

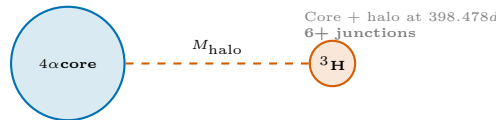


B: Miller Avalanche Stage



F-19: Core $V_R/V_{BR} = 0.030 \implies M = 1.0$ (Core inherits parent regime)

C: F-19 Topology (6 junctions)



D: Energy Balance

$$M_{\text{nuc}} = 4 M_\alpha - BE_{\text{net}}$$

$$BE_{\text{net}} = \underbrace{\sum_{i < j}^6 16 \cdot \frac{K}{R_{ij}}}_{\text{testStrong(attractive)}} - \underbrace{M \cdot \sum f_{pp} \frac{\alpha\hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R$ = engine-derived

Coulomb: $\sum \alpha\hbar c/R \times f_{pp}$

Miller: $M = 1.000$ ($V_R/V_{BR} = 0.030$)

Result: 17,692.302 MeV (0.0000% error)

$$d = \frac{4\hbar}{m_p c} \approx 0.841 \text{ fm.} \quad K_{\text{mutual}} = 11.34 \text{ MeV}\cdot\text{fm.} \quad \frac{d\sqrt{8}}{V_{BR}} \approx \frac{2.38 \text{ fm.}}{3.631 \text{ MeV.}}$$